PAPER_891: SCm Net Energy Buoyancy Regime

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Abstract

Net energy in the SCm vacuum buoyancy regime. $E\_net,SCm = \rho\_SCm(t) \cdot V \cdot c^2 \cdot (2R-1)$. Determines whether the system is in an expansion regime (nebulae, cosmogenesis) or erosion regime (filaments, cavities) based on the buoyancy ratio R = F_{U,Bi}/F_U.

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1. Core Equations

$$E\_net,SCm = \rho\_SCm(t) \cdot V \cdot c^2 \cdot (2R-1)$$

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2. Parameters

Parameter	Default	Description
t	0.0 s	Time parameter
V	1e48 m^3	Volume
F_UBi_over_FU	0.8	Buoyancy-to-field ratio

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3. UQFF Integration

This calculator integrates et_scm_vacuum.py from Session 207 into the CP4 pipeline as class #475.

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